

UserRating MicroService

- Application Name : **MyUserRatingMS**
- Ports : 6500 , 6600 ,6700

Steps to Develop MyUserRatingMS:

A)Setup the Database:

1) Create and open the database

```
drop database jlcratingsdb;  
create database jlcratingsdb;  
use jlcratingsdb;
```

2) Create the tables required

```
drop table myuserratings;  
  
create table myuserratings(  
rating_id int primary key,  
book_id int ,  
user_id char(5) ,  
rating double,  
review char(25)  
);  
  
drop table mybookratings;  
  
create table mybookratings(  
book_id int primary key,  
avg_rating double,  
number_of_searches int  
);
```

3) Insert some sample records

```
insert into myuserratings values(1,101,'U-11',3.5,'Excellent');  
insert into myuserratings values(2,102,'U-12',3.5,'Good');  
insert into myuserratings values(3,101,'U-13',3.5,'Excellent');  
insert into myuserratings values(4,103,'U-11',3.5,'Good');  
insert into myuserratings values(5,104,'U-12',3.5,'Excellent');
```



```
insert into mybookratings values(101,3.5,99);
insert into mybookratings values(102,3.5,99);
insert into mybookratings values(103,3.5,99);
insert into mybookratings values(104,3.5,99);
insert into mybookratings values(105,3.5,99);
insert into mybookratings values(106,3.5,99);
insert into mybookratings values(107,3.5,99);
insert into mybookratings values(108,3.5,99);
insert into mybookratings values(109,3.5,99);
```

4) Select and see the sample records

```
select * from myuserratings;
select * from mybookratings;
```

B)Setup the Boot Project

5) Copy BookPriceMS and Paste as UserRatingMS

6) Remove All except the following

- a) pom.xml
- b) application.properties
- c) JLCBookPriceConfig.java
- d) StartMyBookPriceMS.java

7) Update pom.xml

*** Change Names

```
<groupId>com.jlcindia.userrating</groupId>
<artifactId>UserRatingMS</artifactId>
<version>1.0</version>
<packaging>jar</packaging>

<name>MyUserRatingMS</name>
<url>https://www.jlcindia.com</url>
```

8) Update application.properties

```
spring.application.name=MyUserRatingMS
server.port=6500
spring.datasource.url=jdbc:mysql://localhost:3306/jlcratingsdb
```



9) Rename the Package to com.jlcindia.userratings

10) Rename Spring Bean Configuration Class

*** JLCUserRatingConfig.java

11) Rename BootApplication Starting Class.

*** UserRatingApplication.java

C)Develop the UserRating MicroService Code:

12) Write the JPA Entity

*** BookRating.java

*** UserRating.java

13) Write DAO Layer components

*** UserRatingDAO.java

*** BookRatingDAO.java

14) Write Service Layer components

*** RatingService.java

*** RatingServiceImpl.java

15) Write Controller Layer component

*** RatingController.java

```
@PostMapping("/addUserRating")
```

```
public void addUserRating(@RequestBody UserRating userRating)
```

```
@GetMapping("/userRatings/{userId}")
```

```
public List<UserRating> getUserRatingByUserId(@PathVariable String userId)
```

```
@GetMapping("/bookRatings/{bookId}")
```

```
BookRating getBookRatingByBookId(@PathVariable String bookId)
```

16) Start the Boot Application

17) Open the Swagger UI :

<http://localhost:6500/swagger-ui.html>



18) Test

<http://localhost:6500/userRatings/U-111>
<http://localhost:6500/bookRatings/101>
PUT : <http://localhost:6500/addUserRating>

UserRatingMS: Files required

1. RatingController.java	2. RatingService.java
3. RatingServiceImpl.java	4. UserRatingDAO.java
5. BookRatingDAO.java	6. UserRating.java
7. BookRating.java	8. JLCUserRatingConfig.java
9. UserRatingApplication.java	10. application.properties
11. pom.xml	

1.RatingController.java

```
package com.jlcindia.userratings;

import java.util.List;
import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.web.bind.annotation.*;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@CrossOrigin // CORS
@RestController
public class RatingController {

    static Logger log = LoggerFactory.getLogger(RatingController.class);

    @Autowired
    RatingService ratingService;

    @PutMapping("/addUserRating")
    public void addUserRating(@RequestBody UserRating userRating) {
        log.info("---RatingController---addUserRating()-----");

        ratingService.addUserRating(userRating);
    }
}
```



```
@GetMapping("/userRatings/{userId}")
public List<UserRating> getUserRatingByUserId(@PathVariable String userId) {
    log.info("---RatingController---getUserRatingByUserId()-----");

    return ratingService.getUserRatingByUserId(userId);
}

@GetMapping("/bookRatings/{bookId}")
public BookRating getBookRatingByBookId(@PathVariable String bookId) {
    log.info("---RatingController---getBookRatingByBookId()-----");

    return ratingService.getBookRatingByBookId(Integer.parseInt(bookId));
}
}
```

2. RatingService.java

```
package com.jlcindia.userratings;

import java.util.List;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
public interface RatingService {
    public void addUserRating(UserRating userRating);
    public List<UserRating> getUserRatingByBookId(Integer bookId);
    public List<UserRating> getUserRatingByUserId(String userId);
    public BookRating getBookRatingByBookId(Integer bookId);
}
```

3. RatingServiceImpl.java

```
package com.jlcindia.userratings;

import java.util.List;
import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;
import org.springframework.transaction.annotation.Transactional;
import org.springframework.web.client.RestTemplate;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
```



@Service

@Transactional

```
public class RatingServiceImpl implements RatingService{
```

```
    @Autowired
```

```
    BookRatingDAO bookRatingDAO;
```

```
    @Autowired
```

```
    UserRatingDAO userRatingDAO;
```

```
    @Override
```

```
    public void addUserRating(UserRating userRating) {
```

```
        //1.Add the User Rating
```

```
        userRatingDAO.save(userRating);
```

```
        //2.Calculate the Avg rating for BookId
```

```
        int bookId=userRating.getBookId();//101
```

```
        List<UserRating> ratingList=this.getUserRatingByBookId(bookId);
```

```
        double sumRating=0.0;
```

```
        for(UserRating ur : ratingList) {
```

```
            sumRating=sumRating+ur.getRating();
```

```
        }
```

```
        double avgRating=sumRating/ratingList.size();
```

```
        //3.Update BookRating in UserRatingMS (Local)
```

```
        BookRating bookRating=this.getBookRatingByBookId(bookId);
```

```
        bookRating.setAvgRating(avgRating);
```

```
        bookRatingDAO.save(bookRating);
```

```
        //4.Update BookRating in BookSearchMS(Remote)
```

```
        //Invoking BookRating-MS
```

```
        RestTemplate bookSearchRest=new RestTemplate();
```

```
        String endpoint="http://localhost:8000/updateBookRating";
```

```
        bookSearchRest.put(endpoint, bookRating);
```

```
    }
```

```
    @Override
```

```
    public List<UserRating> getUserRatingByBookId(Integer bookId) {
```

```
        return userRatingDAO.findUserRatingByBookId(bookId);
```

```
    }
```



```
@Override
public List<UserRating> getUserRatingByUserId(String userId) {
    return userRatingDAO.findUserRatingByUserId(userId);
}

@Override
public BookRating getBookRatingByBookId(Integer bookId) {
    BookRating bookRating=bookRatingDAO.findBookRatingByBookId(bookId);
    return bookRating;
}
}
```

4.UserRatingDAO.java

```
package com.jlcindia.userratings;

import java.util.List;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Repository
public interface UserRatingDAO extends JpaRepository<UserRating, Integer> {
    public List<UserRating> findUserRatingByBookId(Integer bookId);
    public List<UserRating> findUserRatingByUserId(String userId);
}
```

5. BookRatingDAO.java

```
package com.jlcindia.userratings;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Repository
public interface BookRatingDAO extends JpaRepository<BookRating, Integer> {
    public BookRating findBookRatingByBookId(Integer bookId);
}
```



6.UserRating.java

```
package com.jlcindia.userratings;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.GeneratedValue;
import javax.persistence.GenerationType;
import javax.persistence.Id;
import javax.persistence.Table;

import io.swagger.annotations.ApiModel;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

@Entity
@Table(name = "myuserratings", schema = "jlcratingsdb")
@ApiModel("UserRating contains details of Rating given by User")
public class UserRating {
    @Id
    @GeneratedValue(strategy = GenerationType.AUTO)
    @Column(name = "rating_id")
    private Integer ratingId;

    @Column(name = "book_id")
    private Integer bookId;

    @Column(name = "user_id")
    private String userId;

    @Column(name = "rating")
    private double rating;

    @Column(name = "review")
    private String review;

    //Constructors
    //Setters and Getters
    //toString
}
```



7. BookRating.java

```
package com.jlcindia.userratings;

import javax.persistence.Column;
import javax.persistence.Entity;
import javax.persistence.Id;
import javax.persistence.Table;

import io.swagger.annotations.ApiModel;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@Entity
@Table(name = "mybookratings", schema = "jlcratingsdb")
@ApiModel("BookRating Contains bookId,avgRating,numberOfSearches")
public class BookRating {

    @Id
    @Column(name = "book_id")
    private Integer bookId;

    @Column(name = "avg_rating")
    private double avgRating;

    @Column(name = "number_of_searches")
    private int numberOfSearches;

    //Constructors
    //Setters and Getters
    //toString
}
```

8.JLCUserRatingConfig.java

```
package com.jlcindia.userratings;

import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.annotation.Bean;
import org.springframework.web.servlet.config.annotation.ResourceHandlerRegistry;
import org.springframework.web.servlet.config.annotation.WebMvcConfigurer;

import com.google.common.base.Predicates;
```



```
import springfox.documentation.builders.PathSelectors;
import springfox.documentation.builders.RequestHandlerSelectors;
import springfox.documentation.service.ApiInfo;
import springfox.documentation.service.Contact;
import springfox.documentation.spi.DocumentationType;
import springfox.documentation.spring.web.plugins.Docket;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */

@SpringBootApplication
public class JLCUserRatingConfig implements WebMvcConfigurer {

    private ApiInfo getApiDetails() {
        return new ApiInfo("JLC BookStore- API",
            "UserRatingMS- API - part of BookStore", "1.0",
            "Free to use ",
            new Contact("Srinivas Dande", "https://www.jlcindia.com", "sri@jlcindia.com"),
            "API Under Free Licence",
            "https://www.jlcindia.com");
    }

    @Bean
    public Docket api() {
        return new Docket(DocumentationType.SWAGGER_2).select().paths(PathSelectors.any())
            .apis(Predicates.not(RequestHandlerSelectors.basePackage("org.springframework.boot")))
            .build()
            .apiInfo(getApiDetails());
    }

    @Override
    public void addResourceHandlers(ResourceHandlerRegistry registry) {
        //enabling swagger-ui
        registry.addResourceHandler("swagger-ui.html").addResourceLocations("classpath:/META-INF/resources/");

        registry.addResourceHandler("/webjars/**").addResourceLocations("classpath:/META-INF/resources/webjars/");
    }
}
```



9. UserRatingApplication.java

```
package com.jlcindia.userratings;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;
import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;

import springfox.documentation.swagger2.annotations.EnableSwagger2;
/*
 * @Author : Srinivas Dande
 * @company : Java Learning Center
 */
@SpringBootApplication
@EnableSwagger2
public class UserRatingApplication {
    static Logger log=LoggerFactory.getLogger(UserRatingApplication.class);

    public static void main(String args[]) {
        log.info(" UserRatingMS - Begin ");
        SpringApplication.run(UserRatingApplication.class, args);
        log.info(" UserRatingMS - End ");
    }
}
```

10. application.properties

```
spring.application.name=MyUserRatingMS

server.port=6500

spring.datasource.url=jdbc:mysql://localhost:3306/jlcratingsdb
spring.datasource.username=root
spring.datasource.password=srinivas

spring.datasource.hikari.connectionTimeout=20000
spring.datasource.hikari.maximumPoolSize=5

logging.level.root=INFO
logging.pattern.console=%-5level %logger{36} - %msg%n

spring.jpa.show-sql=true
spring.jpa.hibernate.ddl-auto=update
```



11. pom.xml

```
<project ...>

<modelVersion>4.0.0</modelVersion>
<groupId>com.jlcindia.userrating</groupId>
<artifactId>UserRatingMS</artifactId>
<version>1.0</version>
<packaging>jar</packaging>

<name>MyUserRatingMS</name>
<url>https://www.jlcindia.com</url>

<parent>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-parent</artifactId>
<version>2.5.3</version>
</parent>

<dependencies>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-web</artifactId>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-starter-data-jpa</artifactId>
</dependency>

<dependency>
<groupId>mysql</groupId>
<artifactId>mysql-connector-java</artifactId>
<version>5.1.6</version>
</dependency>

<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger2</artifactId>
<version>2.4.0</version>
</dependency>
```



```
<dependency>
<groupId>io.springfox</groupId>
<artifactId>springfox-swagger-ui</artifactId>
<version>2.4.0</version>
</dependency>

<dependency>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-devtools</artifactId>
</dependency>

</dependencies>

<build>
<finalName>MyUserRatingMS</finalName>
<plugins>
<plugin>
<groupId>org.springframework.boot</groupId>
<artifactId>spring-boot-maven-plugin</artifactId>
</plugin>
</plugins>
</build>

</project>
```

MicroService Name	Rest Endpoints
MyUserRatingMS	http://localhost:6500/addUserRating
MyUserRatingMS	http://localhost:6500/userRatings/{userId}
MyUserRatingMS	http://localhost:6500/bookRatings/{bookId}